

Portfolio Project 1:

HR Attrition Data Analysis Approach

Tyrone Muhammad, BSDA, CPC

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The primary question for this analysis is: *Why are employees leaving their jobs?* This involves uncovering key patterns or trends related to employee attrition and identifying contributing factors. For example, are specific departments or job roles experiencing higher turnover rates? How do job satisfaction, compensation, and performance metrics influence attrition? Are external factors such as organizational changes impacting employee retention? As *The Big Book of Dashboards* notes, storytelling with data "makes your data and your visualization personal for the user" (Wexler et al., 2017). This analysis aims to create a clear narrative for stakeholders, bridging raw data with actionable insights.

To effectively analyze employee attrition, it is essential to gather the right data:

- **Demographics:** Age, gender, marital status.
- **Job details:** Department, role, years of service, promotions.
- **Compensation metrics:** Salary, bonuses, benefits.
- **Satisfaction scores:** Employee engagement surveys, job satisfaction.
- **Performance metrics:** Productivity reviews, performance ratings.
- **Attrition details:** Reasons for leaving, voluntary vs. involuntary exits.
- **Trends over time:** Attrition rates by month or year.

Without this data, the analysis would be like making a soufflé without eggs; something essential is missing. Once the data is gathered, setting clear parameters ensures the analysis remains focused. Examples include defining a time frame (e.g., last 24 months), segmenting data by department or job role, and identifying "high attrition" as exceeding 10% annual turnover. These steps help avoid "analysis paralysis," where too much data leads to no action—like trying to assemble IKEA furniture without instructions.

The data is sourced from HR Training Data and HRData(1)–(14) files. Using Python, these datasets can be merged using an outer join with `pandas.merge()` and inspected for completeness with methods like `.info()` and `.describe()`. Proper preparation, such as standardizing column names and inspecting for missing or inconsistent data, ensures a smooth analysis. As *Why Data Storytelling Is an Essential Data Science Skill* emphasizes, "presenting the data in a meaningful way is an equally vital part of business intelligence" (Dykes, 2016).

For preprocessing, several steps are essential:

- **Data cleaning:** Remove duplicates and irrelevant columns; standardize text fields to lowercase. After describing the merged dataset, to ensure clarity, the duplicate columns with `_x` and `_y` suffixes are either merged or dropped based on their relevance and completeness."
- **Handling missing data:**
 - Numerical values: Replace missing values with the mean or median.

- Categorical values: Use the mode or create an "NaN" or unknown category.
- **Feature scaling:** Normalize numerical variables, such as salary and years of service, to ensure consistent scaling.
- **Categorical data:**
 - Apply one-hot encoding for variables like department.
 - Use label encoding for ordinal variables like satisfaction levels.

Preprocessing is like preparing a gourmet meal; if the ingredients aren't fresh or properly handled, the end result will disappoint.

Monitoring the reports as data updates is also critical. This can be achieved by automating updates through Python scripts or Power BI refresh functions, ensuring stakeholders always have access to current information. Including a "last updated" timestamp promotes transparency. With new data loads, trends might shift, visualizations may need recalibration, and new insights could emerge, like unexpected spikes in attrition or satisfaction scores needing review.

Creating visual reports makes decision-making easier for stakeholders. For example:

- **Identify trends:** Rising attrition in specific roles or time periods.
- **Compare metrics:** Performance levels vs. attrition rates.
- **Pinpoint issues:** Departments needing intervention based on high turnover.

As *Best Practices in Data Storytelling* highlights, "data analysis is not merely about crunching numbers, but also about selling the outcomes to business stakeholders" (Brown, n.d.). Dynamic visualizations such as interactive dashboards, heatmaps, and bar charts help stakeholders see the story behind the numbers. They not only clarify trends and correlations but also motivate action. A good dashboard sticks in the mind, while a bad one sends stakeholders scrolling through their phones.

Effective visualizations must show trends over time, correlations (e.g., between satisfaction and attrition), and actionable insights (e.g., departments needing attention). Keeping visuals simple and focused is key due to a confused stakeholder is just one question away from asking why their Excel charts don't look as good.

References

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